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Section:BSICS-4K

Lab Task:9

Task:1

Code:

```
File Edit Selection View Go Tools Settings Help
New Open... Save Save As... Undo Redo

1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <unistd.h>
4  #include <signal.h>
5
6
7  void handle_sigusr1(int sig) {
8      if (sig == SIGUSR1) {
9          printf("\nConfiguration reloaded successfully using SIGUSR1\n");
10         fflush(stdout);
11     }
12 }
13
14 int main() {
15
16     if (signal(SIGUSR1, handle_sigusr1) == SIG_ERR)    // this is handling Register signal handler.....
17     {
18         perror("Error setting signal handler");
19         exit(EXIT_FAILURE);
20     }
21
22     printf("Server started with PID: %d\n", getpid());
23
24     // this shoow simulation server work continouslyyy.....
25     while (1) {
26         printf("Server running with current configuration...\n");
27         fflush(stdout);
28         sleep(2);
29     }
30
31     return 0;
32 }
33
```

Output:

now server running with pid 19525

A screenshot of a terminal window titled "Desktop : a.out — Konsole". The window has a menu bar with "File", "Edit", "View", "Bookmarks", "Plugins", "Settings", and "Help". Below the menu bar is a toolbar with icons for "New Tab", "Split View", "Copy", "Paste", and "Find...". The terminal output shows a user prompt "fkhokar@fedora: ~/Desktop\$" followed by the command "./a.out". The output consists of a single line "Server started with PID: 19525" followed by multiple lines of "Server running with current configuration...". A vertical scrollbar is visible on the right side of the terminal area.

After sending signal from other termina...

The screenshot shows a desktop environment with three terminal windows. The leftmost window, titled "Desktop: a.out — Konsole", displays a log of server activity. It shows multiple instances of "Server running with current configuration..." followed by a "Configuration reloaded successfully using SIGUSR1" message. The middle window, titled ": bash — Konsole", shows a terminal prompt where the command "kill -SIGUSR1 19525" has been entered. The bottom window, titled "Unsaved* — Spectacle", is partially visible and appears to be a screenshot tool window.

Task:2

code:

```
New Open... Save Save As... Undo Redo
task1.c task2.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <sys/wait.h>

#define NUM_CHILDREN 3
void sigchld_handler(int sig) {
    int status;
    pid_t pid;

    while ((pid = waitpid(-1, &status, WNOHANG)) > 0) { // this termiante terminate the children
        printf("Child with PID %d terminated.\n", pid);
    }
    fflush(stdout);
}

int main() {
    pid_t children[NUM_CHILDREN];
    int sleep_times[NUM_CHILDREN] = {5, 7, 10}; //this is wait time if chile...

    signal(SIGCHLD, sigchld_handler);

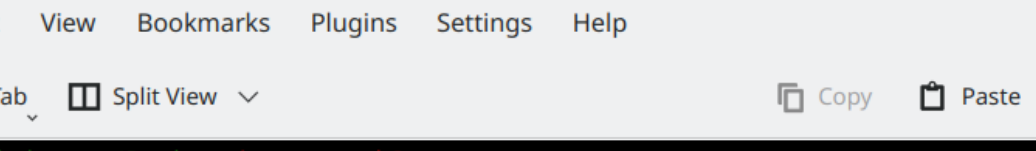
    for (int i = 0; i < NUM_CHILDREN; i++) // this creating children
    {
        pid_t pid = fork();
        if (pid == 0) {
            printf("Child %d started with PID %d\n", i+1, getpid());
            sleep(sleep_times[i]);
            printf("Child %d with PID %d exiting\n", i+1, getpid());
            exit(0);
        } else {
            children[i] = pid;
        }
    }
}
```

```
        children[i] = pid;
    }
}
while (1){
    printf("Server running and monitoring workers...\n");
    fflush(stdout);
    sleep(3);
}

return 0;
}
```

output:

after running



The screenshot shows a terminal window with a dark background. The prompt is `fkhokar@fedora:~/Desktop$`. The user enters `gcc task2.c` and then `./a.out`. The output shows three child processes starting with PIDs 22887, 22888, and 22889, each running a server and monitoring workers. Finally, child 1 (PID 22887) exits, and the parent process (PID 22887) is terminated.

```
fkhokar@fedora:~/Desktop$ gcc task2.c
fkhokar@fedora:~/Desktop$ ./a.out
Child 1 started with PID 22887
Server running and monitoring workers...
Child 2 started with PID 22888
Child 3 started with PID 22889
Server running and monitoring workers...
Child 1 with PID 22887 exiting
Child with PID 22887 terminated.
```

After killing from other terminal

The image displays two terminal windows side-by-side. The left window, titled "Desktop : a.out — Konsole", shows the compilation and execution of a C program named task2.c. The user runs gcc task2.c, followed by ./a.out. The output indicates three child processes were spawned with PIDs 22887, 22888, and 22889. Each child process prints its PID and then terminates, as shown by the "Child with PID ... terminated." messages. The right window, titled "~ : bash — Konsole", shows the user entering the command kill -9 22889, which results in a prompt without further output.

```
Desktop : a.out — Konsole
File Edit View Bookmarks Plugins Settings Help
New Tab Split View Copy Paste Find...
fkhokar@fedora:~/Desktop$ gcc task2.c
fkhokar@fedora:~/Desktop$ ./a.out
Child 1 started with PID 22887
Server running and monitoring workers...
Child 2 started with PID 22888
Child 3 started with PID 22889
Server running and monitoring workers...
Child 1 with PID 22887 exiting
Child with PID 22887 terminated.
Server running and monitoring workers...
Child 2 with PID 22888 exiting
Child with PID 22888 terminated.
Server running and monitoring workers...
Child with PID 22889 terminated.
Server running and monitoring workers...
Server running and monitoring workers...
Server running and monitoring workers...
Server running and monitoring workers...
Server running and monitoring workers...

```

```
~ : bash — Konsole
File Edit View Bookmarks Plugins Settings Help
New Tab Split View Copy Paste Find...
fkhokar@fedora:~$ kill -9 22889
fkhokar@fedora:~$
```